

Arthur Cancellieri Pires

Lead Data Scientist / Python, PyTorch / Anomaly Detection

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Vitoria, Brazil

Professional Summary

Lead Data Scientist with 7+ years building predictive-maintenance ML systems for industrial applications. Expert in Python, PyTorch, and deep learning for time series and anomaly detection, delivering real-time failure predictions at 91% precision and cutting unplanned downtime and maintenance costs. Skilled in scalable ML pipelines, model deployment, and API development (FastAPI, Docker). PhD candidate who leads a 12-person team and ships production ML end to end.

Professional Experience

- Railway Laboratory (Lafer) – partnership with VALE** On-site
LEAD DATA SCIENTIST *August 2024 to Present*
 - Led a 12-person team building end-to-end predictive maintenance ML systems for VALE in Python and PyTorch, taking models from stakeholder requirements to scalable production deployment.
 - Cut equipment failure detection latency from 3 months to real time at 91% precision using deep learning models on multivariate sensor streams.
 - Flagged critical defects weeks before scheduled maintenance, including cases absent from the maintenance database, with unsupervised Transformer autoencoders on time series data.**Skills:** Data Science | Python | PyTorch | PyTorch Lightning | scikit-learn | Deep Learning | Transformers | Autoencoders | Anomaly Detection | Optuna | Deployment | FastAPI | Docker
- Railway Laboratory (Lafer) – partnership with VALE** On-site
SENIOR DATA SCIENTIST *January 2022 to August 2024*
 - Built an automated data-quality framework over the data pipeline that scores the reliability of each data-collection run and filters unreliable sensor data, replacing manual trip-by-trip screening.
 - Developed an unsupervised deep learning autoencoder for sensor anomaly detection, flagging faults missed by rule-based checks at about a 3% false-positive rate and preserving data quality for downstream modeling.
 - Established a systematic methodology to characterize baseline sensor behavior, defining the reference against which the downstream defect-detection models are trained and evaluated.**Skills:** Data Science | Python | SQL | scikit-learn | PyTorch | Deep Learning | Autoencoders | Anomaly Detection | Unsupervised Learning | ETL | Data Quality | Feature Engineering | Signal Processing | PostgreSQL | Optuna
- Railway Laboratory (Lafer) – partnership with VALE** On-site
RESEARCH DATA SCIENTIST *June 2020 to January 2022*
 - Built a Python ETL pipeline that processes 20,000-40,000 sensor records per asset per day, merging multiple raw data sources into a unified analytical dataset powering all downstream machine learning models.
 - Reduced reliance on costly specialized inspections by predicting asset condition at an R2 of about 0.98 with deep learning models on sensor data.**Skills:** Data Science | Python | scikit-learn | PyTorch | PostgreSQL | ETL | Deep Learning | Feature Engineering | Signal Processing | Optuna
- Tribology and Railway Dynamics Laboratory (LabTDF) – partnership with VALE** On-site
MACHINE LEARNING INTERN *Jan 2019 to Jun 2020*
 - Estimated hard-to-measure contact forces at an R2 of about 0.97 from onboard sensors using a simulation-validated ML pipeline, enabling VALE's predictive-maintenance program on instrumented wagons.
 - Optimized wheel/rail wear profiles with genetic algorithms, cutting wear 20% and improving a key safety ratio by approximately 35% without sacrificing fatigue performance.
 - Proved the case for a maintenance threshold change through wear/fatigue analysis, driving adoption into operations and extending service life 29%-50% (29% from the threshold change alone, up to 50% when combined with the optimized wheel profile).**Skills:** Python | MATLAB | Machine Learning | scikit-learn | Optuna | Genetic Algorithms | Optimization | Multibody Simulation

Key Projects

- Industrial Sensor Anomaly Detection API** – Production-style FastAPI service that flags machine faults from vibration sensor streams using per-group residual scoring with stateful multi-window alerting and cooldown; serves models from an MLflow registry with Prometheus metrics, Docker Compose, and GitHub Actions CI. Cut false positives to zero across 29 benchmark scenarios (F1 0.95, recall 0.91), up from a 0.28 F1 baseline. github.com/pirao/AnomalyDetection2026

Education

- State University of Campinas (UNICAMP)** Campinas, Brazil
PH.D. IN MECHANICAL ENGINEERING *Expected: August 2026*
 - Thesis:** A methodology for detecting track defects from instrumented railway vehicle data.
- State University of Campinas (UNICAMP)** Campinas, Brazil
M.SC. IN MECHANICAL ENGINEERING *Jan 2022*
 - Thesis:** Measuring railway track irregularities from instrumented railway vehicle data using machine learning techniques.
- Federal University of Espirito Santo (UFES)** Vitoria, Brazil
B.SC. IN MECHANICAL ENGINEERING *May 2020*
 - Thesis:** Indirect identification of wheel-rail contact forces of an instrumented heavy haul railway vehicle using machine learning.

Languages

Portuguese (Native), English (C1 - Fluent), Italian (A2 - Basic)

Publications & Conferences

SELECTED PUBLICATIONS

- 2024 **Measuring vertical track irregularities from instrumented heavy haul railway vehicle data using machine learning**, Engineering Applications of Artificial Intelligence
- 2022 **Nonintrusive Operator Inference for Chaotic Systems**, IEEE Transactions on Artificial Intelligence
- 2021 **Indirect identification of wheel–rail contact forces of an instrumented heavy haul railway vehicle using machine learning**, Mechanical Systems and Signal Processing

CONFERENCE PRESENTATIONS

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|------|---|-------------------------------|
| 2026 | International Conference on Railway Technology (Railways 2026), Aug. 23–26 (accepted) | <i>Budapest, Hungary</i> |
| 2025 | International Heavy Haul Association (IHHA) Conference, Nov. 17–21 | <i>Colorado Springs, USA</i> |
| 2023 | International Heavy Haul Association (IHHA) Conference, Aug. 27–31 | <i>Rio de Janeiro, Brazil</i> |
| 2023 | Railway Engineering Symposium, May 19–20 | <i>Campinas, Brazil</i> |